QANTUM

A Hybrid Quantum-Classical Financial Intelligence and Portfolio Optimization Engine

Project Proposal | Department of Computer Engineering

Team

Name	Student ID
Jeh Dadina	16010124038
Rishi Dadia	16010124037
Shrish Alva	16010124006

Project guide	Mr. Swapnil Pawar
Co-guide	Dr. Vaibhav Vasani
Thrust area	Intelligent System & Data Processing
Category	Departmental | Internal | No external funding

Abstract

Realistic constraints such as a fixed number of holdings, discrete lots and a budget turn the convex Markowitz problem into a combinatorial one whose search space grows exponentially. QANTUM is a hybrid research platform that tests, with controlled experiments on historical NIFTY 50 data, whether quantum algorithms offer practical value here. Its primary contribution is a QAOA implementation for constrained portfolio selection, formulated as a QUBO and benchmarked against exact MIQP, mean-variance optimisation and simulated annealing on identical instances. A secondary, exploratory module studies Quantum Amplitude Estimation (QAE) for VaR and CVaR against classical Monte Carlo. The project assumes no quantum advantage; it aims to show where these algorithms are and are not competitive.

Problem Statement

Choosing 10 of 50 stocks already gives over 10^{10} candidate portfolios, and tail-risk measures need many Monte Carlo scenarios. No reproducible, end-to-end framework formulates real-market portfolio optimisation and risk estimation for quantum algorithms, runs them alongside strong classical baselines under matched conditions, and evaluates both computational and financial performance. Published QAOA portfolio studies also mostly use US or European equities, and reproducible work on Indian market data is scarce.

Aim

To design, implement and rigorously evaluate a hybrid quantum-classical platform that applies QAOA to constrained portfolio optimisation and explores QAE for financial risk estimation, using historical NIFTY 50 data and benchmarking against established classical methods.

Objectives

ID	Objective	Deliverable
O1	Build a validated data pipeline producing returns, expected returns and shrinkage covariance estimates for NIFTY 50 constituents.	Versioned dataset, data-quality report
O2	Implement classical baselines: continuous MVO, exact MIQP, simulated annealing, and brute force for small N.	Baseline library with certified optima
O3	Derive QUBO/Ising formulations with budget, cardinality and transaction-cost terms, plus penalty calibration.	Formulation module, unit tests
O4	Implement QAOA (standard and CVaR objectives) on Qiskit simulators and IBM hardware.	QAOA engine, depth and optimiser sweeps
O5	Prototype QAE estimation of loss probability, VaR and CVaR, and compare with Monte Carlo at matched query budgets.	Risk module, error-scaling study
O6	Integrate all modules behind a REST API with an experiment store and analytics dashboard.	Running system, reproducible run records
O7	Execute the benchmark protocol and report findings with statistical analysis and stated limitations.	Final report, reproducibility package

Research Questions

RQ1	How effectively can QAOA solve constrained portfolio optimisation problems?
RQ2	How does QAOA solution quality compare with established classical techniques?
RQ3	What are the computational limitations of QAOA as portfolio size increases?
RQ4	Can QAE reduce estimation queries compared with classical Monte Carlo?
RQ5	How do circuit depth, qubit count and noise affect financial optimisation results?
RQ6	Under what conditions, if any, could quantum algorithms give practical advantages in finance?

Proposed System

Module	Role
1. Data acquisition and processing	Clean NIFTY 50 prices; compute returns, volatilities and covariance; record a data hash for traceability.
2. Quantum portfolio optimisation (primary)	Encode the problem as a QUBO, map to Ising, solve with QAOA, decode and repair portfolios; pass the same instance to classical solvers.
3. Quantum risk engine (exploratory)	Iterative QAE for loss probability, VaR and CVaR, compared with Monte Carlo on the same distribution.
4. Market simulation (exploratory)	Scenario generation and stress regimes.
5. Benchmarking environment	Matched instances, budgets, metrics and statistics across all solvers.

Technology Stack

Quantum	Qiskit SDK 2.x, Qiskit Aer, Qiskit IBM Runtime
Computation and backend	Python, NumPy, Pandas, SciPy, scikit-learn, FastAPI, PostgreSQL
Frontend and data	React, TypeScript, Vite, Tailwind CSS, D3.js, yfinance, NSE public data

Methodology and Evaluation

Verification-first: every quantum component is validated against an exact classical reference before it generates results. Instances are random NIFTY 50 sub-universes (N from 4 to 16) with QAOA depth p from 1 to 5, across ideal, noise-model and small hardware runs. Comparisons use matched budgets, and selected portfolios are tested out of sample in walk-forward backtests. Paired results use the Wilcoxon signed-rank test, and negative results are retained.

Metrics: optimality gap, approximation ratio, success probability, feasibility rate; return, volatility, Sharpe ratio, maximum drawdown, VaR, CVaR; qubits, transpiled depth, two-qubit gates, convergence, estimation error versus queries.

Timeline (16 weeks)

Week	Milestone	Evidence
2	M1: Protocol frozen	Research questions, metrics, instance grid documented
4	M2: Dataset released	Versioned NIFTY 50 dataset and estimation pipeline
6	M3: Ground truth available	MIQP, SA and brute force agree on small instances
9	M4: QAOA validated	QAOA reproduces the optimum on small N; sweeps stored
13	M5: System integrated	End-to-end run from dashboard to stored result
16	M6: Final submission	Report, reproducibility package and presentation

Expected Outcomes

A hybrid optimisation engine with a common solver interface; a QAOA implementation with penalty calibration and depth sweeps; a QAE risk prototype; classical benchmarks; an interactive analytics dashboard; a versioned result set with statistical analysis; and a critical evaluation of quantum computing in finance, including negative results.

Limitations

Exact simulation is limited to roughly 25 to 30 qubits. Dense circuits need SWAP routing on sparse hardware, which adds noise. QAOA can face local minima and barren plateaus, and free IBM hardware access is about 10 minutes per 28 days. Instances small enough for today's devices are easy for classical solvers, so the project cannot demonstrate practical quantum advantage and does not attempt to. It characterises how quality and cost scale with N, depth and noise.